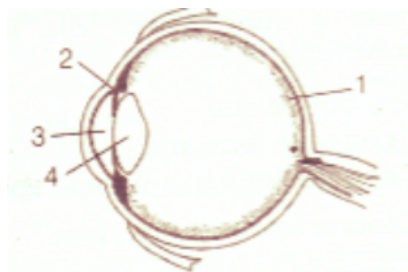


HUMAN EYE -04

Class 10 - Science

1. What is colour blindness ? How can it be cured ? [2]
2. Why cats/ bats are able to see at night ? [2]
3. What change is made in the eye to enable it to focus an object situated at different distances? [2]
4. What is basic cause of colour blindness ? [2]
5. What is the far point and near point of human eye with normal vision? [2]
6. What is meant by power of accommodation of eye? [2]
7. Write the structure of eye lens and state the role of ciliary muscles in the human eye. [2]
8. Why are we not able to see the object clearly for some time when we enter bright light to a room with dim light ? [2]
9. How do we see objects erect when the eye forms a real, inverted image of them on its retina? [2]
10. What is range of vision ? [2]
11. Why chickens come out late in the morning and return early in the evening ? [2]
12. State one main function each of iris, pupil, and cornea. [2]
13. What is accommodation ? [2]
14. a. State one important function of the following parts of the human eye: [2]
 - i. Retina
 - ii. Pupil
 b. State the role of ciliary muscles in focussing objects at varying distances from the eye.
15. Why is a normal eye not able to see clearly the objects placed closer than 25 cm? [2]
16. We are able to see everything with one eye, then why do we have two eyes ? [2]
17. Name the four parts labelled as 1, 2, 3 and 4 in the given figure. [2]



18. Calculate maximum power of accommodation of a person having normal vision. [2]
19. What are the two types of light-sensitive cells on retina of an eye? Which of these cells respond to colour and intensity of light? [2]
20. The limitation of an eye is that the distance from the eye lens to where the image formed is always the same. [2]
How is then it possible for the eye to form focussed images from the objects that are located at varying distances from the eye?

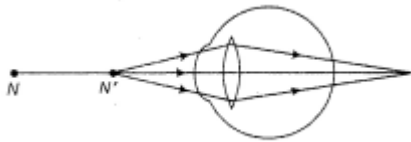
21. How the eye is able to see ? [2]
22. Why the eye lens is not perfectly solid ? [2]
23. Draw a labelled diagram of human eye. What is the significance of the blind spot? [2]
24. How are we able to see nearby and also the distant objects clearly? [2]
25. What is the function of ciliary muscles in human eye ? [2]
26. When is a person said to have developed cataract? How is the vision of such a person restored? [2]
27. A person wants to read a book placed at 20 cm whereas near point of his eye is 30 cm. Calculate the power of the lens required. [2]
28. Does myopia or hypermetropia imply that the eye has partially lost its accommodation ability? If not, what cause these defects of vision? [2]
29. Reena was diagnosed with a short sight problem. She was advised to wear spectacles continuously which made her feel low about her beauty. What would you suggest Reena other than wearing spectacles? [2]
30. A student uses spectacles of focal length -2.5 m. [2]
 - i. Name the defect of vision he is suffering from.
 - ii. Which lens is used for the correction of this defect?
 - iii. List two main causes of developing this defect.
 - iv. Compute the power of this lens.
31. A myopic person can see things clearly only when they lie between 10 cm and 100 cm from the eye. Which lens will enable him to see the Moon clearly? [2]
32. Make a diagram to show how hypermetropia is corrected. The near point of a hypermetropic eye is 0.8 m. What is the power of the lens required to correct this defect? Assume that the near point of the normal eye is 25 cm. [2]
33. The far point of a myopic person is 120 cm in front of the eye. What is the nature and power of the lens required to correct the problem? [2]
34. List two differences in the characteristic properties of the virtual images formed by the two types of spherical lenses (concave and convex). How are these characteristics of the two lenses used in the correction of the two common defects of vision namely myopia and hypermetropia? [2]
35. Why does a person suffering from hypermetropia prefer to remove his spectacles while looking at distant objects? Explain. [2]
36.
 - i. List two causes of hypermetropia. [2]
 - ii. Draw ray diagrams showing (i) a hypermetropic eye and (ii) its correction using a suitable optical device.
37. In a certain murder investigation, it was important to discover whether the victim was long-sighted or short-sighted. How could a detective decide by examining his spectacles? [2]
38. John has undergone all medication including wearing spectacles, lens and laser treatment. None of the treatment helped him to recover his eyesight. What would be your suggestion to overcome his problem in his eyesight? [2]
39. A person is suffering from an eye defect in which the far point of the eye is nearer than infinity. Identify the defect. List two main causes of this defect. [2]

Draw a ray diagram to show how this defect is corrected by using a suitable lens.
40. When do we consider a student sitting in the class to be myopic? List two causes of this defect. Explain using a ray diagram how this defect of eye can be corrected. [2]
41. Why in old ages the eye sight becomes foggy? How can this defect may be removed? [2]
42. Ram can read a book clearly only if he holds it at an arm's length from him. Name the defect of vision if he is an old man. [2]

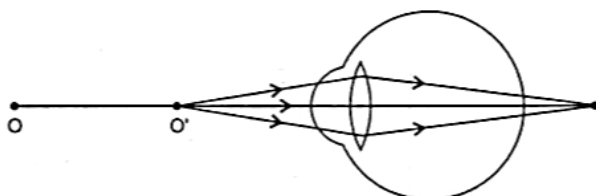
43. What is name of defect of eye due to loss of elasticity of eye-lens ? How is it corrected ? [2]
44. Draw ray diagrams each showing [2]
 - i. myopic eye and
 - ii. hypermetropic eye.
45. When we increase the distance of an object from the eye, what happens to the image distance in the eye? [2]
46. What type of lens we wear if we are suffering from myopia ? [2]
47. Zia has been suggested by doctor to avoid blue light to her eyes which is emitted from all digital items. But in today's world, the day doesn't finish off without the usage of digital items. What would be the preventive measures given by you to your friend? [2]
48. List two main causes of a person developing near sightedness. Show with the help of a ray diagram how this defect can be corrected. [2]
49. A person with myopic eye cannot see objects beyond 1.2 m distinctly. What should be the type of the corrective lens used to restore proper vision? [2]
50.
 - i. Why does the power to see near objects as well as far off objects clearly diminish with age? Name the defects that are likely to arise in eye in such a condition. [2]
 - ii. The far point of a myopic eye is 60 cm. Find the focal length of the lens used to correct it.
51. Your friend's daughter is facing a lot of eye problems. Though she restricts her daughter from watching TV and mobiles, the kid is not able to recover from the eyesight problems effectively. What would be your suggestion to increase the sight of your friend's daughter? [2]
52. Babin was diagnosed with a long sight problem. He was advised to wear lens but he was worried that wearing lens should be given an utmost care in his busy schedule. What would you suggest Babin other than usage of lens? [2]
53. State reasons for Myopia. With the help of ray diagrams, show the [2]
 - a. image formation by a myopic eye, and
 - b. correction of myopia using an appropriate lens.
54. A student needs spectacles of power -0.5 D for the correction of his vision. [2]
 - a. Name the defect in vision the student is suffering from.
 - b. Find the nature and focal length of the corrective lens.
 - c. List two causes of this defect.
55. A doctor has prescribed a corrective lens of power - 1.2 D to a person suffering from defect of vision. [2]
 - i. Identify the defect from which he is suffering.
 - ii. Find the focal length of the lens.
 - iii. Is the prescribed lens diverging or converging? Show the nature of this lens with the help of a ray diagram.
56. A person cannot see distinctly the object placed beyond 5 m from his eyes. Name the defect of vision the person is suffering from. Draw a ray diagram to illustrate this defect. List its two possible causes. Name the lens used for the correction of this defect. [2]
57. What would be your suggestion to your grandfather who is undergoing the eye defects both hypermetropia and myopia? [2]
58. The far point of a myopic person is 80 cm in front of the eye. What is the nature and power of the lens required to correct the problem. [2]
59. The near point of a hypermetropic person is 50 cm. What should be the focal length and power of the lens used [2]

by him if he wants to read a newspaper placed 20 cm. away ?

60. What is astigmatism ? How is it corrected ? [2]
61. Parthiv with normal near point (25 cm) reads a book with small print using a magnifying glass, a thin convex lens of focal length 5 cm. What are the closest and farthest distances at which he can read the book viewing through the magnifying glass? [2]
62. Richie's grandmother was affected by cataract in which the crystalline lens of his mother become so hard cloudy substance. Though the doctor advised opting for traditional cataract surgery, Richie needs something better in which his grandma shouldn't suffer a lot but it should be carefully and precisely done. As a young medico student, what would you opt for? [2]
63. Study the diagram given below and answer the questions that follow: [2]



- Which defect of vision is represented in this case? Give reason for your answer.
 - What could be the two causes of this defect?
 - With the help of a diagram show how this defect can be corrected by the use of a suitable lens?
64. What is presbyopia ? How can it be corrected ? [2]
65. i. A person is suffering from both myopia and hypermetropia. [2]
- What kind of lenses can correct this defect?
 - How are these lenses prepared?
- ii. A person needs a lens of power + 3D for correcting his near vision and -3D for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects.
66. A student has difficulty reading the blackboard while sitting in the last row. What could be the defect the child is suffering from? How can it be corrected? [2]
67. Ahmed cannot distinguish between red and green colour. Why? Does he have normal vision? [2]
68. A person is unable to see clearly a poster fixed on a distant wall. He however sees it clearly when standing at a distance of about 2 m from the wall. [2]
- Draw ray diagram to show the formation of image by his eye lens when he is far away from the wall.
 - List two possible reasons of this defect of vision.
 - Draw ray diagram to show the correction of this defect using appropriate lens.
69. Far point of a myopic eye is 20 cm. What type of lens should be wear to see a picture on T.V. placed 2.5 m away from him ? [2]
70. The figure shows a human eye suffering form a certain type of defect in the eye. [2]
- Name the defect of eye.
 - What type of lens is needed to rectify the above defect?
 - Draw a diagram to show how the above defect has been removed by using the above lens.



71. A person may suffer from both myopia and hypermetropia defects. [2]

- i. What is this condition called?
 - ii. When does it happen?
 - iii. Name the type of lens often required by the persons suffering from this defect. Draw labelled diagram of such lenses.
72. A person needs a lens of power -5.5 dioptres for correcting his distant vision. For correcting his near vision he needs a lens of power $+1.5$ dioptre. What is the focal length of the lens required for correcting
- i. distant vision
 - ii. near vision?
73. What happens when elasticity of the lens is reduced to zero ? [2]
74. A student sitting at the back of the classroom cannot read clearly the letters written on the blackboard. What advice will a doctor give to her? Draw the ray diagram for the correction of this defect. [2]
75. Draw a labelled diagram of a common type of bifocal lens. State the function of each of its parts. Name the defect for the correction of which it is used and state its main cause. [2]